

Exam 2

2a. Python

b. Python

c. The Poisson distribution appears roughly normal with a large peak reaching 0.1 in density. It has a low spread, indicating a lower variance. The Normal distribution appears roughly normal with a peak at 20 reaching a density of 0.02. It has a much larger spread due to the standard deviation being 20.

d. Poisson: $\bar{x} = 20.004$ $s^2 = 20.3976$
Normal: $\bar{x} = 19.9979$ $s^2 = 397.2301$

e. The normal approximation to a Poisson distribution improves as λ becomes large because the mean and variance equal λ , so as λ grows the distribution becomes more bell-shaped and symmetric. This also follows the Central Limit Theorem but based off of λ .

f. The approximation is worse because λ is smaller. The Poisson distribution is originally skewed right for smaller λ and this is shown in this case. There is also a lower variance.

g. Python

h. Python

i. The Gamma distribution is roughly normal with a large peak reaching 0.04 density. It has a low spread indicating a lower variance. The Normal distribution is roughly normal with a much lower peak of 0.005. It has a much larger spread due to the variance being larger matching the bell-shape.

Exam 2

- 2j. The approximation becomes less accurate because the shape parameter, α , is much lower, leading the distribution to be skewed to the right as compared to normal.
- K. The features of a distribution that make a normal approximation more effective is a larger sample size for the Central Limit Theorem, a larger variance, and larger values for variables that effect shape.

Exam 2

39. Python

b. Python

c. Python

d. Studytime: $\bar{x} = 1.93$ $s^2 = 0.69$ Final Grade: $\bar{x} = 11.91$ $s^2 = 10.44$

e. Sample cov = 0.67

f. Sample corr = 0.25

g. A correlation of 0.25 means that study time and the final grade have a positive and weak linear relationship.

h. This relationship should be interpreted carefully because a linear relationship can not really be created due to the ordinal nature of study time. Due to this, the data is 4 vertical lines that is not indicative of each student's exact study time and resulting score.

i. A joint distribution shows how two variables work together, while the marginals only show how variable behaves on its own. It shows the relationship between two variables.